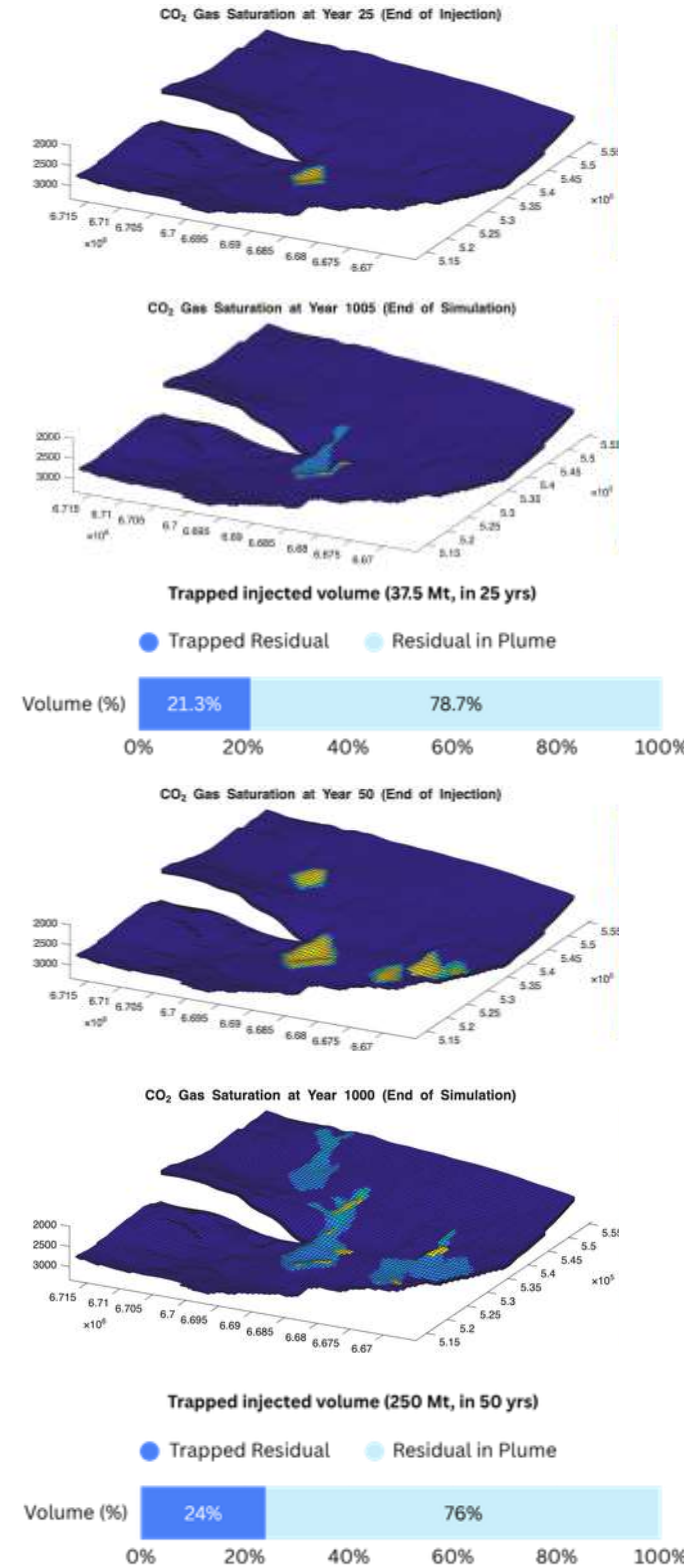


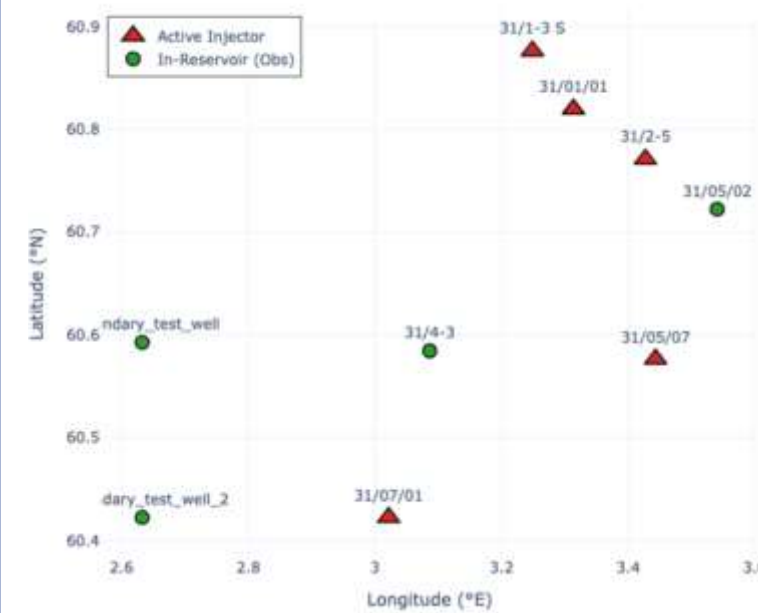


Application of Reservoir and Production Surveillance for Optimized Reservoir Management in Conventional or Unconventional Fields (Johansen Field Injection Optimisation)

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Johansen Formation — In-Reservoir Wells



SIMULATION

Before:

- ❑ Only one well was injecting @ 1.5Mt/yr (as per Phase - I) with 6 surveillance wells
- ❑ Total injection plan 37.5 Mt in 25 yrs.

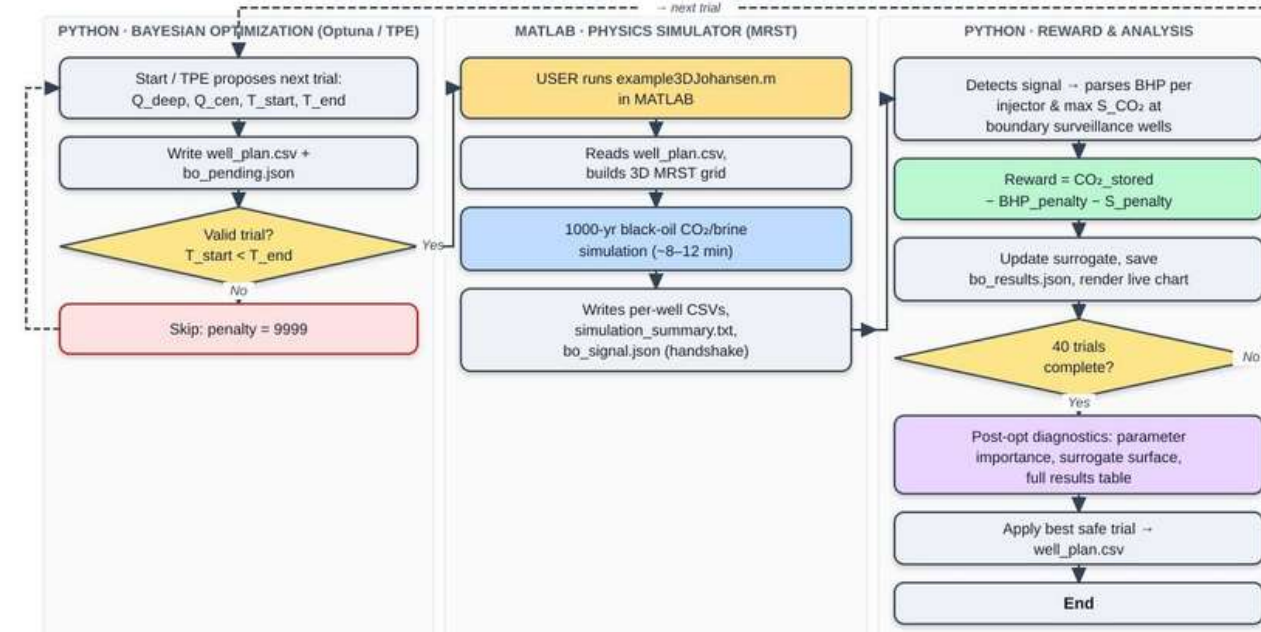
Proposition:

- ❑ 5 injector and 2 surveillance, 6.875Mt/yr
- ❑ Total injection plan 250 Mt in 50 yrs.

Constraints:

- ❑ Overhead Caprock Fracture Pressure (335 bar)
- ❑ Saturation of CO₂ should be < 0.15 (critical CO₂ saturation) for Johansen, at boundary

Johansen CO₂ BO System — Three-Layer Handshake Workflow



OPTIMISATION

- ❑ Boundary surveillance data directly gates the next injection proposal — monitoring becomes decision-making.
- ❑ Containment failure, pressure breach are penalised, encoding geophysical irreversibility into the reward function.
- ❑ Two boundary sentinels on opposing plume pathways ensure no unsafe strategy escapes constraint detection.
- ❑ Staggered inter-cluster injection timing emerges as an optimised pressure management strategy — never prescribed, always discovered.

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